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TP 01 : Exploration des Rôles et Fonctions du SE

Exercise 1 : Identification and Classification

Task 1.1 :

```
alaghrib@alaghrib-Precision-7550:~$ uname -a
Linux alaghrib-Precision-7550 6.14.0-37-generic #37~24.04.1-Ubuntu SMP PREEMPT_DYNAMIC Thu Nov 20 10:25:38 UTC 2 x86_64 x86_64 x86_64 GNU/Linux
alaghrib@alaghrib-Precision-7550:~$
```

FIGURE 1 – Task 1.1 Screenshot

- Kernel name : **Linux**

- Architecture : x86-64 the system is **64 bits**.

Which means the virtual address space is 2^{64} , far larger than 32-bit systems which are limited only **4GB**.

Task 1.2 :

```
alaghrib@alaghrib-Precision-7550:~$ who
alaghrib tty7      2026-02-24 18:45 (:0)
alaghrib@alaghrib-Precision-7550:~$ top

top - 22:47:47 up 4:02, 1 user, load average: 0.60, 0.51, 0.60
Tasks: 329 total, 1 running, 327 sleeping, 0 stopped, 1 zombie
%cpu(s): 3.9 us, 1.2 sy, 0.0 ni, 94.7 id, 0.3 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 15626.0 total, 10654.7 free, 3505.8 used, 2529.2 buff/cache
MiB Swap: 6676.0 total, 6676.0 free, 0.0 used, 12120.2 avail Mem

  PID USER   PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+  COMMAND
1988 alaghrib 20   0 12.9g 777516 315212 S 16.2  4.9 21:36.36 firefox-bin
11480 alaghrib 20   0 3534624 125456 89708 S 11.6  0.8 0:02.25 Isolated Web Co
1521 alaghrib 20   0 4606836 217464 121580 S 10.3  1.4 6:11.25 cinnamon
1052 root      20   0 1021792 153512 103760 S 7.3  1.0 3:29.10 Xorg
2617 alaghrib 20   0 19.9g 379468 170588 S 5.6  2.4 7:07.92 Isolated Web Co
8514 alaghrib 20   0 3570244 151612 101432 S 5.3  0.9 0:04.35 Isolated Web Co
1108 alaghrib 9 -11 116352 15208 9236 S 1.3  0.1 0:00.56 pipewire
2653 alaghrib 20   0 3532720 96464 77420 S 1.3  0.6 0:13.86 Isolated Servic
11561 alaghrib 20   0 3542208 122764 90304 S 1.0  0.8 0:01.19 Isolated Web Co
794 root      20   0 259076 15352 12836 S 0.7  0.1 0:10.97 touchegg
11357 alaghrib 20   0 14544 6000 3740 R 0.7  0.0 0:00.57 top
17 root      20   0 0 0 0 S 0.3  0.0 0:08.54 ksoftirqd/0
790 root      20   0 424808 11716 10580 S 0.3  0.1 0:25.05 thermald
1230 alaghrib 20   0 388888 11700 6008 S 0.3  0.1 0:14.83 ibus-daemon
1335 alaghrib 20   0 230372 6384 5528 S 0.3  0.0 0:00.09 dconf-service
3173 root     0 -20 0 0 0 I 0.3  0.0 0:08.48 kworker/u49:0-i915_flip
7734 root      20   0 0 0 0 I 0.3  0.0 0:12.56 kworker/0:2-i915-unordered
7757 alaghrib 20   0 8269476 392744 161476 S 0.3  2.5 9:31.50 Isolated Web Co
11326 alaghrib 20   0 690740 44412 35044 S 0.3  0.3 0:01.53 gnome-terminal-
11593 alaghrib 20   0 3503912 70496 55804 S 0.3  0.4 0:00.16 Web Content
11608 alaghrib 20   0 3503912 70480 55780 S 0.3  0.4 0:00.19 Web Content
1 root      20   0 22864 13996 9568 S 0.0  0.1 0:02.46 systemd
2 root      20   0 0 0 0 S 0.0  0.0 0:00.02 kthreadd
3 root      20   0 0 0 0 S 0.0  0.0 0:00.00 pool_workqueue_release
4 root      0 -20 0 0 0 I 0.0  0.0 0:00.00 kworker/R-rcu_gp
5 root      0 -20 0 0 0 I 0.0  0.0 0:00.00 kworker/R-sync_wq
6 root      0 -20 0 0 0 I 0.0  0.0 0:00.00 kworker/R-kvfree_rcu_reclaim
7 root      0 -20 0 0 0 I 0.0  0.0 0:00.00 kworker/R-slub_flushwq
8 root      0 -20 0 0 0 I 0.0  0.0 0:00.00 kworker/R-netns
10 root     0 -20 0 0 0 I 0.0  0.0 0:00.00 kworker/0:0H-events_highpri
13 root     0 -20 0 0 0 I 0.0  0.0 0:00.00 kworker/R-mm_percpu_wq
14 root     0 -20 0 0 0 I 0.0  0.0 0:00.00 rcu_tasks_kthread
```

FIGURE 2 – Task 1.2 Screenshot

- The 'Who' command shows only one connected user (alaghrib), but the system supports multiple user , The system is Multi-Users .
- The 'top' command shows 329 tasks running (1 running ,327 sleeping ,0 sttoped ,1 zombie),proving the OS uses **preemptive multitasking**.

Exercise 2 : Observation of Resource management

Task 2.1 (Process Management) :

- System process : systemd(PID 1, user root)
- User process : firefox-bin(PID 1988, user alaghrib)
- The most CPU-consuming process is **firefox-bin** with 16.2 CPU .

Task 2.2 (Memory Management) :

- Total RAM : 15026 MB (Used RAM : 3505.8 MB)
- Swap total : 6676 MB (Used swap : 0.0 MB)
- Swap acts as an extension of RAM on disk . When RAM is full , the OS moves inactive pages to Swap, which is the concept of **virtual memory** allowing the system to run more processes than physical RAM allows.

Exercise 3 : Storage Abstraction (Files)

Task 3.1 :

- The root directory / contains all system files and directories.
- The /etc directory stores all system configuration files such as hostname, passwd, network, ssh, sudo.conf, and many others .
- The ls /home command shows that only one user exists on this system : **alaghrib** .

```
alaghrib@alaghrib-Precision-7550:/$ ls /
bin          boot  dev  home  lib64  lost+found  mnt  proc  run /sbin.usr-is-merged  sys  usr
bin.usr-is-merged  cdrom  etc  lib  lib.usr-is-merged  media  opt  root  sbin  srv  var

alaghrib@alaghrib-Precision-7550:/$ ls /etc
adduser.conf  cron.weekly  groff  libao.conf  netconfig  rc3.d  sysctl.conf
adjtime       cron.yearly  group  libaudit.conf  netplan  rc4.d  sysctl.d
alsa          cryptsetup-initramfs  grub.d  libblockdev  networkd-dispatcher  rc5.d  systemd
alternatives  cups         gshadow  libbl-3      networkd-dispatcher  rc6.d  terminfo
anacrontab    dbus-1       gss      libpaper.d   NetworkManager  rcS.d  thermal
apache2       dbusconf.conf  gss      libpcre2     networkd-dispatcher  request-key.conf  timeshift
app.conf      debconf.conf  gss      libpcre2     networkd-dispatcher  request-key.d  timezone
app           debconf.conf  gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
apparmor      debconf.conf  gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
apparmor.d    debconf.conf  gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
apt           debconf.conf  gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
avahi         debconf.conf  gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
bash.bashrc   debconf.conf  gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
bash_completion  dhcpcd.conf  gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
bindresport.blacklist  dictionaries-common  gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
bluetooth     dirmngr.conf  gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
brlapi.key    e2scrub.conf  gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
brltty        emac          gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
brltty.conf   environment   gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
ca-certificates  etherotypes  gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
casper.conf   fonts         gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
catdocrc     fpm           gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
chattr        fuse.conf     gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
cifs-utils    gpg           gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
colord        gpg.conf     gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
console-setup  gpg.conf     gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
crackmap      gpg.conf     gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
credstore     gpg.conf     gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
credstore.encrypted  gpg.conf     gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
cron.d        gpg.conf     gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
cron.daily    gpg.conf     gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
cron.hourly   gpg.conf     gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
cron.monthly  gpg.conf     gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
cronstab      gpg.conf     gss      libpcre2     networkd-dispatcher  request-key.d  timeshift
alaghrib@alaghrib-Precision-7550:/$ ls /home
alaghrib  lost+found
```

FIGURE 3 – Task 3.1 Screenshot

Task 3.2 (Security) :

```
alaghrib@alaghrib-Precision-7550:/etc$ ls -l
total 1280
-rw-r--r-- 1 root root 3444 Jul 5 2023 adduser.conf
-rw-r--r-- 1 root root 18 Feb 7 16:42 adjtime
drwxr-xr-x 3 root root 4096 Jan 8 19:32 alsa
drwxr-xr-x 2 root root 12288 Feb 17 14:34 alternatives
-rw-r--r-- 1 root root 335 Apr 8 2024 anacrontab
drwxr-xr-x 3 root root 4096 Feb 11 23:22 apache2
-rw-r--r-- 1 root root 433 Apr 8 2024 apg.conf
drwxr-xr-x 5 root root 4096 Jan 8 19:32 apm
drwxr-xr-x 3 root root 4096 Jan 8 19:32 apparmor
drwxr-xr-x 7 root root 4096 Feb 7 16:44 apparmor.d
drwxr-xr-x 8 root root 4096 Feb 7 16:44 apt
drwxr-xr-x 3 root root 4096 Jan 8 19:32 avahi
-rw-r--r-- 1 root root 2319 Mar 31 2024 bash.bashrc
-rw-r--r-- 1 root root 45 Jan 24 2020 bash_completion
drwxr-xr-x 2 root root 4096 Feb 11 23:22 bash_completion.d
-rw-r--r-- 1 root root 367 Aug 2 2022 bindresvport.blacklist
drwxr-xr-x 2 root root 4096 Apr 19 2024 binfmt.d
drwxr-xr-x 2 root root 4096 Jan 8 19:32 bluetooth
-rw-r----- 1 root root 33 Jan 8 19:32 brlapi.key
drwxr-xr-x 7 root root 4096 Jan 8 19:32 brltty
-rw-r--r-- 1 root root 30571 Mar 31 2024 brltty.conf
drwxr-xr-x 3 root root 4096 Jan 8 19:32 ca-certificates
-rw-r--r-- 1 root root 6288 Jan 8 19:32 ca-certificates.conf
-rw-r--r-- 1 root root 437 Apr 18 2024 casper.conf
-rw-r--r-- 1 root root 119 Jan 11 2022 catdocrc
drwxr-xr-x 2 root dip 4096 Jan 8 19:32 chatscripts
drwxr-xr-x 2 root root 4096 Jan 8 19:32 cifs-utils
drwxr-xr-x 2 colord colord 4096 Feb 7 16:26 colord
drwxr-xr-x 2 root root 4096 Feb 7 16:42 console-setup
drwxr-xr-x 2 root root 4096 Jan 8 19:32 cracklib
drwx----- 2 root root 4096 Apr 19 2024 credstore
drwx----- 2 root root 4096 Apr 19 2024 credstore.encrypted
drwxr-xr-x 2 root root 4096 Jan 8 19:32 cron.d
drwxr-xr-x 2 root root 4096 Jan 8 19:32 cron.daily
drwxr-xr-x 2 root root 4096 Jan 8 19:32 cron.hourly
drwxr-xr-x 2 root root 4096 Jan 8 19:32 cron.monthly
-rw-r--r-- 1 root root 1136 Mar 31 2024 crontab
drwxr-xr-x 2 root root 4096 Jan 8 19:32 cron.weekly
drwxr-xr-x 2 root root 4096 Jan 8 19:32 cron.yearly
drwxr-xr-x 2 root root 4096 Jan 8 19:32 cryptsetup-initramfs
drwxr-xr-x 5 root lp 4096 Feb 25 00:00 cups
```

FIGURE 4 – Task 3.2 Screenshot

- Every file has an owner and permissions (r=read, w=write, x=execute) for owner, group, and others. When a user tries to modify a file, the OS checks if they have w permission (write access) — if not, access is immediately denied with Permission denied. In this system .